

LIMBERT, Antarctica

WMO: 892570

Lat: 75.914S

Lon: 59.264W

Elev: 131

StdP: 14.63

Time Zone: -3.00 (W03)

Period: 98-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	-56.9	-53.0	-65.1	0.1	-56.8	-61.3	0.1	-53.0	58.5	-8.9	57.5	-9.8	6.6	220	1.335

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
12	11.1	32.4	30.3	30.4	28.8	28.6	27.2	29.6	30.7	27.9	29.0	26.3	27.3	8.4	20

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
28.1	22.1	30.1	26.2	20.3	28.5	24.5	18.7	27.1	10.9	30.9	10.2	29.1	9.5	27.5	42.8

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 33.6	28.1	23.9	-61.4	39.1	2.4	4.6	-63.1	42.5	-64.5	45.2	-65.8	47.8	-67.6	51.2		
(5)			N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	-9.1	20.1	7.0	-9.8	-19.5	-22.6	-25.4	-28.4	-26.8	-20.1	-11.5	8.1	20.7		
	DBStd	20.85	6.05	9.64	12.31	11.54	13.06	13.26	13.20	13.63	12.71	11.50	8.77	6.17		
	HDD50	21571	927	1204	1852	2084	2252	2263	2431	2381	2103	1907	1257	910		
	HDD65	27046	1392	1624	2317	2534	2717	2713	2896	2846	2553	2372	1707	1375		
	CDD50	0	0	0	0	0	0	0	0	0	0	0	0	0		
	CDD65	0	0	0	0	0	0	0	0	0	0	0	0	0		
	CDH74	0	0	0	0	0	0	0	0	0	0	0	0	0		
	CDH80	0	0	0	0	0	0	0	0	0	0	0	0	0		
	WSAvg	9.8	9.0	10.7	10.6	10.1	9.1	8.0	9.0	9.6	9.9	11.4	10.7	9.5		
	PrecAvg	16.4	0.9	0.9	1.1	1.1	1.5	1.5	1.7	2.0	2.2	1.6	1.2	0.8		
(15) Precipitation	PrecMax	23.2	1.7	1.8	2.3	2.0	3.9	3.0	4.8	3.2	3.9	3.5	2.9	1.9		
	PrecMin	10.8	0.1	0.2	0.2	0.3	0.3	0.3	0.4	0.4	0.9	0.2	0.1	0.1		
	PrecStd	3.1	0.4	0.4	0.5	0.6	0.7	0.7	0.8	0.7	0.6	0.7	0.6	0.4		
	PrecStd	3.1	0.4	0.4	0.5	0.6	0.7	0.7	0.8	0.7	0.6	0.7	0.6	0.4		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	37.6	30.6	24.9	13.8	17.2	16.0	16.7	19.6	13.3	22.7	30.3	36.1		
		MCWB	35.7	28.9	24.3	13.2	17.2	15.5	16.7	19.6	12.5	21.1	28.1	32.4		
	2%	DB	33.2	27.4	19.3	6.8	9.5	9.0	5.1	9.4	9.1	15.2	26.1	32.2		
		MCWB	31.3	26.4	18.9	6.4	8.9	8.2	4.6	9.0	8.7	14.7	24.8	29.9		
	5%	DB	30.8	25.0	14.3	1.6	3.6	0.9	-0.8	0.4	3.8	10.5	23.7	30.5		
		MCWB	29.3	24.5	13.9	1.1	3.2	0.4	-1.4	0.0	3.2	9.8	22.7	28.8		
	10%	DB	28.6	22.2	8.5	-2.9	-3.6	-6.6	-8.6	-6.3	-1.4	6.2	21.5	28.9		
		MCWB	27.4	21.4	7.8	-3.4	-4.0	-7.0	-9.0	-6.7	-1.8	5.6	20.6	27.5		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	32.9	28.6	23.7	11.8	12.1	17.9	9.6	19.4	12.1	19.2	27.7	31.1		
		MCDB	34.5	29.3	24.1	12.2	12.2	18.8	10.0	19.4	12.4	19.5	29.2	32.7		
	2%	WB	29.9	26.3	19.7	7.0	8.1	11.8	2.3	10.5	7.0	14.3	24.9	29.5		
		MCDB	30.9	26.9	20.3	7.4	8.6	12.4	2.8	11.0	7.7	14.7	26.1	30.8		
	5%	WB	27.7	24.5	15.8	1.3	3.4	0.8	-4.1	1.1	0.3	9.8	22.7	28.3		
		MCDB	28.7	25.1	16.2	1.7	3.9	1.3	-3.8	1.4	0.7	10.3	23.5	29.5		
	10%	WB	25.7	21.5	9.2	-3.2	-3.0	-6.9	-10.6	-5.4	-3.8	5.9	20.9	26.8		
		MCDB	26.7	22.3	9.8	-2.8	-2.6	-6.5	-10.2	-4.9	-3.4	6.4	21.8	28.1		
(35) Mean Daily Temperature Range	5% DB	MDBR	11.0	14.3	14.3	15.0	15.9	15.8	16.9	16.6	16.4	16.0	14.0	11.1		
		MCDBR	12.2	14.2	19.1	20.8	20.7	23.2	27.8	23.8	22.7	17.8	13.7	12.5		
		MCWBR	9.8	13.0	17.5	20.2	17.9	22.9	28.4	22.0	24.1	16.3	12.5	10.6		
		MCDBR	9.7	13.9	18.6	20.1	17.9	23.2	27.3	22.2	23.0	17.1	13.1	11.0		
	5% WB	MCWBR	8.9	13.4	18.4	20.1	17.8	23.0	27.0	22.0	22.6	16.6	12.2	9.8		
(40) Clear-Sky Solar Irradiance	taub		0.247	0.232	0.194	0.120	N/A	N/A	N/A	0.126	0.194	0.215	0.230	0.247		
	taud		2.222	2.232	2.197	1.899	N/A	N/A	N/A	1.860	2.133	2.189	2.214	2.203		
	Ebn at Noon		305	284	245	92	0	0	0	75	237	290	311	312		
	Edh at Noon		33	26	16	3	0	0	0	3	17	27	33	36		
(44) All-Sky Solar Radiation	RadAvg		2387	1495	561	67	0	0	0	18	325	1126	2179	2726		
	RadStd		77	67	28	4	0	0	0	1	16	32	66	74		

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (0 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page